

CP317 - Software Engineering

Project Title: Car Rental System

Deliverable: Milestone 2

Group ID: Team # 28

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Car Rental System

Product Development Plan and Initial Design

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1. Expanded Product Backlog

The product roadmap and its requirements form the basis of the product backlog, which is a prioritized list of tasks for the development team. In order to direct the development lifecycle, it has been extended to incorporate story points, priority levels, and sprint assignments. With improvements in Sprint 3, this framework guarantees that we provide a Minimum Viable Product (MVP) at the conclusion of Sprint 2.

The foundation of both Scrum and Agile approaches is a product backlog. It breaks down the specific tasks of the project and reveals the priorities in addition to capturing all the desired features and enhancements.

Story points are estimated by using Fibonacci sequence (1, 2, 3, 5, 8), a common practice in Agile development to represent the relative effort, complexity, and uncertainty of each task, rather than absolute time. This method acknowledges the inherent uncertainty in estimation and encourages a team-based, relative sizing approach.

Story ID	User Story	Priority	Story Points	Sprint	Status
AUTH-1	As a new user, I want to create an account with my email and a secure password so that I can access the platform's features.	High	3	1	Not Started
AUTH-2	As a registered user, I want to log in and out securely so that my personal information and booking history are protected.	High	2	1	Not Started
UI-1	As a customer, I want to search for available cars by date and location on the homepage so that I can quickly see my options.	High	3	1	Not Started
UI-2	As a customer, I want to see a list of available cars with key details (image, model, price) after searching so that I can compare my choices.	High	5	1	Not Started
UI-3	As a customer, I want to view a detailed page for a specific car with more images, full specifications, and total cost so that I can make an informed decision.	High	5	2	Not Started
RES-1	As a customer, I want to add a selected vehicle to a shopping cart so that I can finalize my booking without losing my selection.	High	3	2	Not Started
RES-2	As a customer, I want to complete a reservation by providing payment details (simulated) so that my booking is confirmed.	High	5	2	Not Started
USER-1	As a registered user, I want a personal dashboard to view my current and past reservations so that I can manage my rental history.	Medium	5	2	Not Started

ADMIN-1	As an administrator, I want to log into a separate admin portal so that I can manage the system's operations.	High	3	2	Not Started
ADMIN-2	As an administrator, I want to add, view, and update vehicle details in the fleet inventory so that the car listings are always accurate.	High	8	2	Not Started
UI-4	As a customer, I want to filter search results by car category and price range so that I can narrow down my options more effectively.	Medium	5	3	Not Started
REP-1	As an administrator, I want to view a dashboard with key performance indicators (KPIs) like fleet utilization so that I can monitor business health.	Medium	8	3	Not Started
PAY-1	As a developer, I need to integrate a simulated payment gateway to process transactions and finalize bookings.	High	5	3	Not Started

2. Requirements Refinement

A Software Requirements Specification (SRS) document is essential for minimizing revising and setting precise goals. It links requirements to design, coding, and testing and establishes success criteria. This process ensures all features can be validated and helps prevent scope creep.

2.1. Narrative Description

Our Car Rental System is a feature-rich online platform created to digitize and optimize the whole car rental procedure. Customers and administrators of rental companies are the two main user groups that the system will cater to. In addition to viewing thorough information for each vehicle, customers will be able to create a secure account and conduct thorough searches for available automobiles based on dates, car type, and price. The basic functionality allows them to make and manage reservations through a personalized dashboard. Using a dedicated interface, administrators may utilize the technology to manage the whole fleet of vehicles, adding new models, updating data, and designating vehicles as requiring maintenance. They will be able to provide useful information on fleet usage and business success in addition to having a real-time overview of all reservations. The system acts as a safe, efficient, and consolidated center for all rental operations.

2.2. Key Functional Requirements

Functional requirements define the specific behaviors or functions of the system. They describe what the system should do and are essential for demonstrating its internal workings. These requirements are used to explain each subsystem's purpose.

- **User Management:** includes safe ways to register, log in, and log out.
- **Vehicle Search:** A dynamic search feature that let the use choose the vehicles based on rental dates, categories and price range
- **Vehicle Details:** A page for each and every vehicle that will display vehicle's images, specifications, and availability status.
- **Reservation System:** Function to make, view, and cancel bookings/ reservations. Includes a review booking to hold selections before checkout.
- **Customer Dashboard:** A personal interface for users to manage their profile and view rental history.
- **Administrative Dashboard:** Administrators can monitor all system functions from the administrative dashboard.
- **Inventory Management:** Resources to help administrators create, read, update, and deactivate (CRUD) the fleet of vehicles.
- **Reporting & Analytics:** Generation of key business reports, such as fleet utilization and booking trends.
- **Payment Simulation:** Integration of a simulated payment gateway to finalize transactions

2.3. Important Non-Functional Requirements

Often referred to as quality attributes, non-functional requirements (NFRs) outline a system's performance rather than its capabilities. Both a strong, secure product and a satisfying user experience depend on them. Data integrity, speed, security, and dependability are all covered by NFRs.



Non-functional requirements define the quality attributes of a software system

- **Usability:** In both contemporary desktop and mobile browsers, the user interface needs to be responsive, easy to use, and consistent. The ability of a new user to finish a booking in less than five minutes is a crucial indicator. According to best standards for contemporary web design, the design should be simple and put the user experience first. Good design enhances the overall experience.
- **Security:** Security is really essential. A trustworthy method, such as bcrypt, must be used to hash each users' password. To ensure data security while in transit, all data communication must be conducted via HTTPS. Encrypting personally identifiable information (PII) is necessary when it is at rest. The system ought to adhere to best practices such as those described in the OWASP Top Ten to help lessen typical weaknesses.
- **Performance:** To ensure a smooth user experience and prevent user drop-off, key pages such as the Homepage and Search Results must load in under 3 seconds under normal load conditions. This includes optimizing images and database queries.
- **Accessibility:** Our app will be easy to use for everyone (considering people with disabilities), This basically means we're working on things like screen reader support, making sure you can navigate with just a keyboard, and that the colors have enough contrast. It's not just the best practice, in a lot of places, it's also the law.
- **Reliability:** The system must demonstrate high availability (e.g., 99.9% uptime) and include robust error handling to prevent crashes from unexpected user input or system faults. This ensures the platform is consistently available and trustworthy for users.

3. Ethical Consideration

Integrating ethical principles into the software development lifecycle is not an afterthought but a core requirement. It involves a commitment to creating software that is fair, transparent, and respects user privacy.

3.1. Data Privacy & Security

Our mitigation strategy is operationalized through specific backlog items and non-functional requirements. User stories AUTH-1 and AUTH-2 explicitly require password hashing. Data in transit is protected by requiring HTTPS to be used during deployment. A crucial step is also putting Role-Based Access Control (RBAC) into practice, as is intended for storylines ADMIN-1 and ADMIN-2. By following the least privilege principle, RBAC makes sure that only authorized administrative staff have access to sensitive customer data.. RBAC provides a structured approach to managing access permissions, which is fundamental for compliance with regulations like GDPR. The concept of 'privacy by design', mandated by GDPR, is integrated from the start.

3.2 Accessibility

We're designing this to be accessible from start to end. That means we're using clear colors, a logical structure, and navigation that will be easy for anyone to use. Every feature will work seamlessly with screen readers and a keyboard, making sure we meet important accessibility standards. Ultimately, The website will give better user experience and will be usable for people with disabilities also.

3.3 Algorithmic Fairness

The logic will only be based on objective, documented factors: rental period, vehicle category, and booking lead time. This will assure openness and fairness, particularly if dynamic pricing is implemented in the future. We shall specifically refrain from feeding any private user information, like demographics or past location, into pricing algorithms. This increases user trust and stops discriminatory actions.. Research on algorithmic bias highlights the risk of discrimination based on protected attributes, a pitfall we are determined to avoid. These rules are documented in the "Conversation" notes for relevant backlog items to ensure transparency and prevent hidden biases.

4. Initial Design Sketches (Wireframes)

A digital product's wireframe can be assumed as its blueprint or skeleton, it defines the arrangement and positioning of components prior to the start of visual design (mockup) and development. They are an essential tool for bringing teams together around requirements and concentrating discussions on key features and user flow. These low-fidelity drawings serve as the application's framework, facilitating quick feedback and iterations.

4.1. Homepage / Search Landing Page

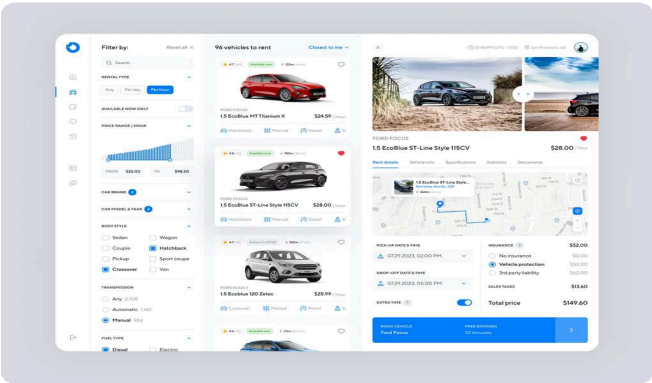
The webpage is made to encourage quick action. It has a clear navigation bar (Logo, Home, Login/Signup) and a simple, minimalistic design. The main feature is a huge "Search" button and a visible search form with easy-to-use fields for Pick-up Location and Pick-up & Return Dates. The main user task—locating a car—is given priority in this design. To engage customers who might be browsing without a clear plan, a section for featured autos or special deals can be placed beneath the main search module.



Wireframe concept for the homepage, emphasizing a direct and user-friendly search interface

4.2. Search Results Page

The webpage is made to encourage quick action. It has a clear navigation bar (Logo, Home, Login/Signup) and a simple, minimalistic design. The main feature is a huge "Search" button and a visible search form with easy-to-use fields for Pick-up Location and Pick-up & Return Dates. The main user task—locating a car—is given priority in this design. To engage customers who might be browsing without a clear plan, a section for featured autos or special deals can be placed beneath the main search module.



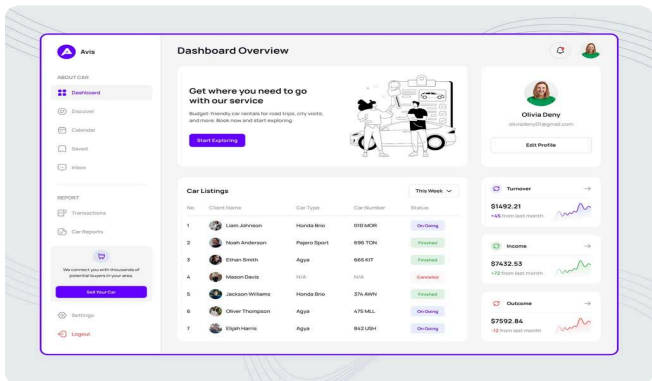
Layout for the search results page, featuring a filter panel and a detailed list of available vehicles

4.3. Vehicle Details Page

The visitor is directed to the Vehicle Details Page after choosing a vehicle. The purpose of this screen is to instill confidence and offer thorough information. It has a sizable picture gallery that allows you to view the car from various perspectives. Features and all specifications are covered in the thorough description that appears beneath the gallery. A clear cost breakdown for the chosen rental dates that eliminates any hidden costs is an essential part. A big "Add to Cart" or "Reserve Now" button at the end of the page provides a compelling and obvious path to conversion.

4.4. User Dashboard

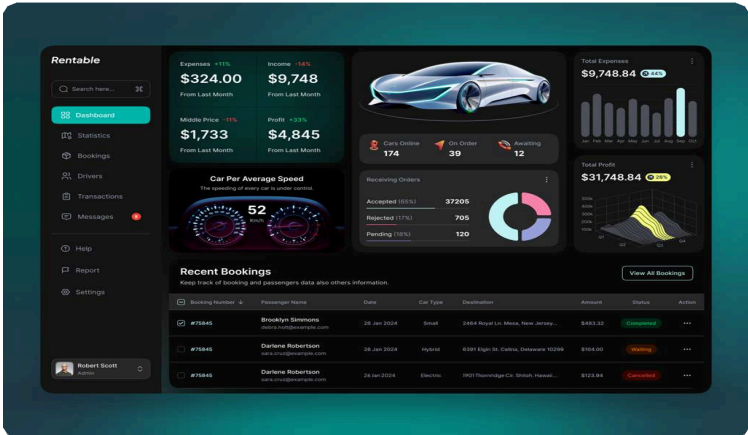
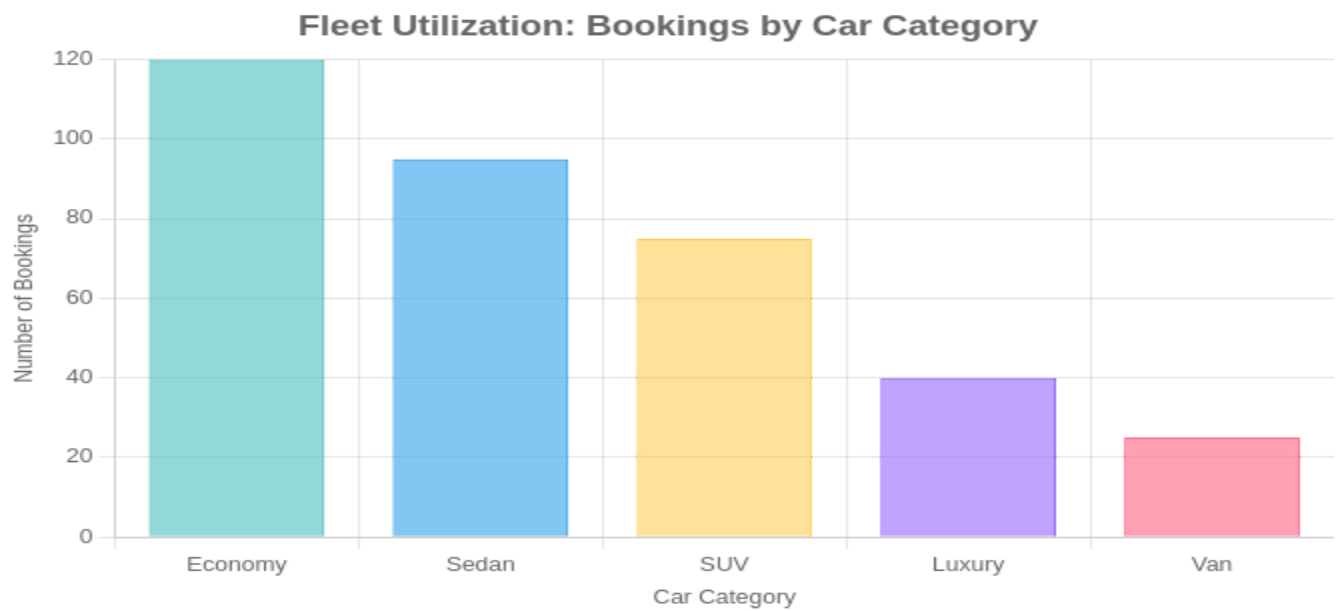
Users can customize their experience through the User Dashboard, which is organized into several sections, starting with an introduction. Key features include “Current Reservations,” where users can view their active bookings, check details, or cancel them; “My Profile,” which lets users update their personal information; and “Rental History,” which provides an organized record of all past reservations.



A clean user dashboard design showing a profile summary and a list of current car listings or reservations

4.5. Admin Dashboard

The Admin Dashboard serves as the command center for rental operators. It’s designed to highlight the company’s overall performance through key performance indicators (KPIs) such as “Cars Rented Today,” “Total Available Cars,” and “Revenue.” The dashboard also provides quick access to essential management tools, including “View All Bookings,” which displays a detailed list of all reservations for operational monitoring, and “Manage Fleet,” where administrators can view, edit, or deactivate vehicles as needed.



A comprehensive admin dashboard concept displaying key financial metrics and recent booking activity